

Avant Max Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Avant Protocol on Avant Max. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated August 28, 2025, was delivered to Avant Protocol, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Avant Max code provided by Avant Protocol. The findings consist of 2 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Avant Max
Repository	Avant-Contracts-Max
Commit	1e2f0ad6ca38...
Fix Commit	cc074dbd6aeb...
Audit Timeline	Aug 22nd - Aug 26th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	2	0	2
Informational	7	4	3
Gas Optimizations	5	5	0
Total	14	9	5

6 Audit Scope

The scope of this audit was limited to:

```
script/Deployment.s.sol
src/AddressesWhitelist.sol
src/PriceStorage.sol
src/RequestsManager.sol
src/SimpleToken.sol
```

Additionally we were asked to investigate the on-chain state of the following deployed contracts:

avBTCx (Avalanche)

```
SimpleToken: 0xa7C10C510df4B1702E1F36451dd29D7C3EDC760C
  impl: 0xf7cf101e9c3d6035a9f832a0c02efbce56f7dfc7
PriceStorage: 0x40B418cF176731089B2537D027A14c78a86F2166
  impl: 0x0c66fcd3eac84a04a198f2ffe00ca64f3c9272af
RequestsManager: 0x5F0AEf33a03Bf0028fC46ddDD4a86ee3d29E2972
AddressesWhitelist: 0x89245A4Bd8948713Fd5f6DA7c84CF6D2b76BEd7B
avUSDx Token: 0xDd1cDFA52E7D8474d434cd016fd346701db6B3B9
  impl: 0xaefa3ffe45781680d5ad99627b7eb9d79192b29a
avUSDx Pricing Contract: 0x7b4e8103bdDD5bcA79513Fda22892BEE53bA9777
  impl: 0x0ef7be66249217e7588ad9277806080472357923
avUSDx Requests Manager: 0x4C129d3aA27272211D151CA39a0a01E4C16Fc887
```

avETHx (Ethereum)

```
SimpleToken: 0x2E8b7190eE84E7AC757Ddff42Ba14d4EAe24B865
  impl: 0x1411f6d18d4e0015ee2dc22d4f7b6893dedbdbe1
PriceStorage: 0x985b5eF57dBE19B4EE1eD43b9AdEB1A61f2f6f23
  impl: 0x511c025ed8c04e363d398c21d165563adddbee99
RequestsManager: 0xb8CE3a51233299F1aBAdc84e78820cca457F413a
AddressesWhitelist: 0x570C0EB59655407D137B207E60cE3aB103CC1c4d
```

7 On-Chain State Analysis

The deployed contracts on Ethereum and Avalanche listed in the Audit Scope section were essentially the same contracts as those audited, differing in their manner of deployment.

Cyfrin verified this by:

- using block explorers to analyze the deployed code
- using cast to list the available public functions of the deployed contracts (`cast interface`) and to find their implementation addresses (`cast implementation`), then `cast interface` on the implementations
- using block explorers to analyze deployment-related transactions

For example:

```
cast interface 0x2E8b7190eE84E7AC757Ddff42Ba14d4EAe24B865
cast interface 0xa7C10C510df4B1702E1F36451dd29D7C3EDC760C --chain avalanche
cast implementation 0x2E8b7190eE84E7AC757Ddff42Ba14d4EAe24B865
cast implementation --rpc-url https://api.avax.network/ext/bc/C/rpc
↪  0xa7C10C510df4B1702E1F36451dd29D7C3EDC760C
```

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

The protocol implements a token minting and burning system that bridges external assets into wrapped versions through a request-based mechanism. At its core, `RequestsManager` serves as the central orchestrator, accepting deposits of whitelisted tokens from providers who request minting of the wrapped token. These requests enter a pending state until they are completed with the final mint amount, which may be greater but not lesser than the requested amount due to exchange rates.

The system works bidirectionally: users can also burn their wrapped tokens to request redemption of the underlying assets, following the same request-complete pattern. The treasury address receives deposited tokens from completed mints and provides tokens for burns, keeping funds separate from the operational contracts.

The architecture employs several supporting contracts:

- `SimpleToken` provides the ERC20 implementation with mint/burn
- `AddressesWhitelist` maintains an optional KYC layer, allowing participation to be restricted to verified addresses when regulatory compliance is required
- `PriceStorage` stores prices

10 Attestation

Cyfrin attests that it performed a security review of the Avant Max code provided by Avant Protocol.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-08-28-cyfrin-avant-max-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Avant Protocol has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=ce4f9bdf76bbc5254915549ff527ce48ba6c3ba9ac31dacafafcee654f316a46
F=2025-08-28-cyfrin-avant-max-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.